Black Scholes Merton Option Pricing

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3. Solution of the Black-Scholes Equation
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Basic setup

- The main idea of Black and Scholes lies in the construction of a riskless portfolio taking positions in risk-free asset, option, and the underlying stock.
- Black and Scholes transformed the option pricing problem into the task of solving a (parabolic) PDE with a final condition.
- The most important application of the Ito calculus, in financial mathematics is the pricing of options.

Assumptions under BSM model

- The options are European
- No dividends are paid out during the life of the option
- Market movements cannot be predicted (or market is efficient)
- Nil transactions cost
- The risk-free rate and volatility of the underlying asset are known and constant
- Asset price follows a lognormal distribution i.e. log-returns are normally distributed.

Steps in deriving Black-Scholes equation

- Assume that the asset price $S(t)$ follows the geometric Brownian motion,

$$dS(t) = \mu S(t)dt + \sigma S(t)dB(t).$$

- Let $V = V(S(t), t)$ denote the value of an option (or a contingent claim) that is sufficiently smooth, namely, its second-order derivatives with respect to S and first-order derivative with respect to t are continuous.
- For an Ito process $dX(t) = P(t)dB(t) + Q(t)dt$, the Ito formula is given by

$$\begin{aligned} df(t, X(t)) &= f_t(t, X(t))dt + f_x(t, X(t))P(t)dB(t) \\ &\quad + f_x(t, X(t))Q(t)d(t) + \frac{1}{2} f_{xx}(t, X(t))P^2(t)dt. \end{aligned}$$

- Then, it immediately follows from the Ito lemma that

$$dV = \left(\mu S \frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dB.$$

Payoffs from equity and call option

- Since both stochastic processes S and V are driven by the same Wiener process (Brownian motion) B, the stochastic term, $\sigma S \frac{\partial V}{\partial S} dB$, can be eliminated by constructing a portfolio that consists of the option and the underlying asset.
- Let P be the wealth of the portfolio that consists of one short position in option with value V and Δ units of the underlying asset with the price S.
- The value of the portfolio at time t is given by

$$P = -V + \Delta S,$$

  where P_0 is the initial capital of the portfolio.
- Therefore, the infinitesimal change in the portfolio becomes

$$\begin{aligned} dP &= -dV + \Delta dS \\ &= -\left[\left(\mu S \frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right) dt + \sigma S \frac{\partial V}{\partial S} dB\right] + \Delta \left(\mu S dt + \sigma S dB\right) \\ &= -\left(\mu S \left(\Delta - \frac{\partial V}{\partial S}\right) + \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right) dt + \left(-\frac{\partial V}{\partial S} + \Delta\right) \sigma S dB. \end{aligned}$$

- By choosing $\Delta = \frac{\partial V}{\partial S}$, we can hedge the portfolio.
- Thus the infinitesimal change dP of the portfolio within the time interval dt is

$$dP = -\left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right) dt \tag{1}$$

- Further, in infinitesimal time dt the portfolio capital will earn the risk-free rate that is given by

$$dP = rPdt = r(-V + \Delta S)dt \tag{2}$$

$$= -rV + rS\frac{\partial V}{\partial S} \tag{3}$$

- Under no arbitrage principle, the two investment should give the same infinitesimal change dP. Thus

$$-rV + rS\frac{\partial V}{\partial S} = -\left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right) \tag{4}$$

- Hence,

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0. \tag{5}$$

- This is the celebrated Black-Scholes equation for European options.

Solution of the Black-Scholes Equation

Definition (Fourier transform)

For a function f, the Fourier transform is given by

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} e^{-i\omega x} f(x) dx, \quad -\infty < \omega < \infty.$$

Definition (Inverse Fourier transform)

The inverse Fourier transform is given by

$$\hat{f}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega x} \hat{f}(\omega) d\omega.$$

Fourier Transform of Gaussian Function

Suppose $f(x) = e^{-ax^2/2}$, $a > 0$. Then

$$f'(x) = -axf(x),$$

taking Fourier transform in both side yields

$$\begin{aligned} i\omega\hat{f}(\omega) &= -ai\frac{d}{d\omega}\hat{f}(\omega) \\ &\implies \frac{d}{d\omega}\hat{f}(\omega) = -\frac{\omega}{a}\hat{f}(\omega) \\ &\implies f'(\omega) = Ae^{-\frac{\omega^2}{2a}}. \end{aligned}$$

To calculate A, put $\omega = 0$, which implies

$$A = \hat{f}(0) = \int_{-\infty}^{\infty} e^{-\frac{ax^2}{2}}\,dx = \sqrt{\frac{2\pi}{a}}.$$

Thus

$$\mathcal{F}\left(e^{-\frac{ax^2}{2}}\right) = \sqrt{\frac{2\pi}{a}}e^{-\frac{\omega^2}{2a}}.$$

Standard heat equation

The standard heat equation is

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad -\infty < x < \infty, \ t > 0,$$
$$u(x, 0) = f(x).$$

Taking Fourier transform in both hand side implies

$$\frac{d}{dt}\hat{u}(\omega, t) = c^2 (i\omega)^2 \hat{u}(\omega, t)$$
$$\implies \hat{u}(\omega, t) = A(\omega) e^{-c^2 \omega^2 t},$$

where $A(\omega)$ is a constant depending on ω.

We have $u(x,0) = f(x)$, which implies $\hat{u}(\omega,0) = \hat{f}(\omega)$, leading to $A(\omega) = \hat{f}(\omega)$. Hence

$$\hat{u}(\omega,t) = \hat{f}(\omega)e^{-c^2\omega^2 t}.$$

Using convolution property of Fourier transform, we have

$$\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g) \implies \mathcal{F}^{-1}(\hat{f}\hat{g}) = f * g.$$

Thus

$$u(x,t) = \mathcal{F}^{-1}(u(\omega,t)) = \frac{1}{2c\sqrt{\pi t}}\int_{-\infty}^{\infty} f(s)e^{-\frac{(x-s)^2}{4c^2 t}}\,ds.$$

Here, we have used that

$$\mathcal{F}^{-1}(e^{-c^2\omega^2 t}) = \frac{1}{2c\sqrt{\pi t}}e^{-\frac{x^2}{4c^2 t}}.$$

The Black-Scholes equation is given by

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0, \tag{6}$$

where $V(0,t) = 0$ and $V(S,T) = \max\{S - K, 0\}$. This is a parabolic PDE.

Classification of Second Order PDE

Suppose the linear second order PDE has the form

$$A\frac{\partial^2 u}{\partial x^2} + 2B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + D = 0,$$

then this PDE is called

- Parabolic if $B^2 - AC = 0$
- Elliptic if $B^2 - AC < 0$
- Hyperbolic if $B^2 - AC > 0$.

Make the following change of variables in Black-Scholes PDE

- $t = T - \frac{\tau}{\sigma^2/2}$, which gives $\tau = \frac{\sigma^2}{2}(T - t)$.
- $S = Ke^x$, which gives $x = \log(\frac{S}{K})$.
- Also put $V(S, t) = Kv(x, \tau)$. Thus $V(S, T) = Kv(x, 0)$, since $T = T - \frac{\tau}{\sigma^2/2} \implies \tau = 0$.
- $\frac{\partial V}{\partial t} = \frac{\partial V}{\partial \tau}\frac{d\tau}{dt} = K\frac{\partial v}{\partial \tau}(-\frac{\sigma^2}{2})$.
- $\frac{\partial V}{\partial S} = \frac{\partial V}{\partial x}\frac{dx}{dS} = K\frac{\partial v}{\partial x}(\frac{1}{S})$.
- Further,

$$\frac{\partial^2 V}{\partial S^2} = \frac{\partial}{\partial S}\left(\frac{\partial V}{\partial S}\right) = \frac{\partial}{\partial S}\left(K\frac{\partial v}{\partial x}\frac{1}{S}\right) = K\frac{\partial v}{\partial x}\left(-\frac{1}{S^2}\right) + K\frac{\partial}{\partial S}\left(\frac{\partial v}{\partial x}\right)\frac{1}{S}$$
$$= K\frac{\partial v}{\partial x}\left(-\frac{1}{S^2}\right) + K\frac{\partial}{\partial x}\left(\frac{\partial v}{\partial x}\right)\frac{dx}{dS}\frac{1}{S} = K\frac{\partial v}{\partial x}\left(-\frac{1}{S^2}\right) + K\frac{\partial^2 v}{\partial x^2}\left(\frac{1}{S^2}\right).$$

- Also, $V(S, T) = \max\{S - K, 0\} = K\max\{e^x - 1, 0\}$.
- But $V(S, T) = Kv(x, 0)$ which implies $v(x, 0) = \max\{e^x - 1, 0\}$.

Substituting these values in BS equation, yields

$$\frac{\partial v}{\partial \tau} = \frac{\partial^2 v}{\partial x^2} + (p-1)\frac{\partial v}{\partial x} - pv = 0,$$

where $p = \frac{r}{\sigma^2/2}$ and $v(x,0) = \max\{e^x - 1, 0\}$.

Observations

- In this equation, we have one parameter p.
- The equation is defined on the interval $-\infty < x < \infty$, since $x = \log(S/K), S, K > 0$.
- The equation has constant coefficients.

Next substitute

$$v(x,\tau) = e^{\alpha x+\beta\tau}u(x,\tau)$$

where α and β will be chosen appropriately. We have

- $v_\tau = e^{\alpha x+\beta\tau}u_\tau + \beta e^{\alpha x+\beta\tau}u$
- $v_x = e^{\alpha x+\beta\tau}u_x + \alpha e^{\alpha x+\beta\tau}u$
- $v_{xx} = e^{\alpha x+\beta\tau}u_{xx} + 2\alpha e^{\alpha x+\beta\tau}u_x + \alpha^2 e^{\alpha x+\beta\tau}u$
- Putting in constant coefficient equation yields,

$$\beta u + u_\tau = \alpha^2 + 2\alpha u_x + u_{xx} + (p-1)(\alpha u + u_x) - pu.$$

The last equation reduces to

$$u_\tau = u_{xx} + (2\alpha + p - 1)u_x + (\alpha^2 + (p-1)\alpha - p - \beta)u.$$

- Choose $2\alpha + p - 1 = 0$ and $\alpha^2 + (p-1)\alpha - p - \beta = 0$.
- This give $\alpha = \frac{1-p}{2}$ and $\beta = -\frac{1}{4}(p+1)^2$.
- The PDE reduces to

  $$u_\tau = u_{xx},$$

  which is the standard heat equation.
- The initial condition will change to

  $$\begin{aligned} u(x,0) = e^{-\alpha x} v(x,0) &= e^{\frac{(p-1)}{2}x} \max\{e^x - 1, 0\} \\ &= \max\{e^{\frac{(p+1)}{2}x} - e^{\frac{(p-1)}{2}x}, 0\}. \end{aligned}$$

- Note that $u(x,0) > 0$ when $x > 0$ and $u(x,0) = 0$ for $x \leq 0$.

- The solution of the heat equation is

$$u(x,\tau) = \frac{1}{2\sqrt{\pi\tau}} \int_{-\infty}^{\infty} u_0(s) e^{-\frac{(x-s)^2}{4t}} ds.$$

- Taking $z = \frac{(s-x)}{\sqrt{2\tau}}$, which implies $dz = -\frac{1}{\sqrt{2\tau}} dx$, so the integration becomes

$$u(x,\tau) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u_0(z\sqrt{2\tau} + x) e^{-\frac{z^2}{2}} dz.$$

- Here $u_0 > 0$ for $z > -\frac{x}{\sqrt{2\tau}}$.
- The solution reduces to

$$\begin{aligned} u(x,\tau) &= \frac{1}{\sqrt{2\pi}} \int_{-x/\sqrt{2\tau}}^{\infty} e^{\frac{k+1}{2}(x+z\sqrt{2\tau})} e^{-\frac{z^2}{2}} dz \\ &\quad - \frac{1}{\sqrt{2\pi}} \int_{-x/\sqrt{2\tau}}^{\infty} e^{\frac{k-1}{2}(x+z\sqrt{2\tau})} e^{-\frac{z^2}{2}} dz \\ &= I_1 - I_2. \end{aligned}$$

- We have

$$I_1 = \frac{1}{\sqrt{2\pi}} \int_{-x/\sqrt{2\tau}}^{\infty} e^{\frac{k+1}{2}(x+z\sqrt{2\tau})} e^{-\frac{z^2}{2}} dz$$
$$= \frac{1}{\sqrt{2\pi}} e^{\frac{(k+1)}{2}x+\tau\frac{(k+1)^2}{4}}) \int_{-x/\sqrt{2\tau}}^{\infty} e^{-\frac{1}{2}(z-\sqrt{\frac{\tau}{2}}(k+1))^2} dz.$$

- Puting $y = z - \sqrt{\frac{\tau}{2}}(k+1))^2$, leads to

$$I_1 = \frac{1}{\sqrt{2\pi}} e^{\frac{(k+1)}{2}x+\tau\frac{(k+1)^2}{4}} \int_{-x/\sqrt{2\tau}-\sqrt{\frac{\tau}{2}}(k+1)}^{\infty} e^{-\frac{y^2}{2}} dy$$
$$= e^{\frac{(k+1)}{2}x+\tau\frac{(k+1)^2}{4}} \Phi(d_1),$$

where $d_1 = x/\sqrt{2\tau} + \sqrt{\frac{\tau}{2}}(k+1)$ and Φ is the standard normal CDF.

- Similarly we have

$$I_2 = \frac{1}{\sqrt{2\pi}} e^{\frac{(k-1)}{2}x+\tau\frac{(k-1)^2}{4}} \int_{-x/\sqrt{2\tau}-\sqrt{\frac{\tau}{2}}(k-1)}^{\infty} e^{-\frac{y^2}{2}} dy$$
$$= e^{\frac{(k-1)}{2}x+\tau\frac{(k-1)^2}{4}} \Phi(d_2),$$

  where $d_2 = x/\sqrt{2\tau} + \sqrt{\frac{\tau}{2}}(k-1)$ and Φ is the standard normal CDF.
- Thus the transformed heat equation has solution

$$u(x,\tau) = e^{\frac{(k+1)}{2}x+\tau\frac{(k+1)^2}{4}} \Phi(d_1) - e^{\frac{(k-1)}{2}x+\tau\frac{(k-1)^2}{4}} \Phi(d_2).$$

Back Substitution

Make the following substitutions

- $v(x,\tau) = e^{-\frac{(k-1)}{2}x - \frac{(k+1)^2}{4}\tau} u(x,\tau)$
- $x = \log(S/K)$
- $\tau = \frac{\sigma^2}{2}(T-t)$
- $V(S,T) = Kv(x,\tau)$.

Closed Form Solution of the BSM PDE at Time t

The price of a European call option is given by

$$V(S, K, r, \sigma, T, t) = S\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2),$$

where Φ is standard normal CDF and d_1 and d_2 are defined as

$$d_1 = \frac{\log(S/K) + (r + \sigma^2/2)(T-t)}{\sigma\sqrt{(T-t)}}$$

and

$$d_2 = \frac{\log(S/K) + (r - \sigma^2/2)(T-t)}{\sigma\sqrt{(T-t)}}.$$

- S is the underlying asset price
- K is the strike price
- r is the risk-free rate
- T is the time to maturity
- σ is the volatility of the underlying asset.
- t is the time at which option price is calculated.

Price at time 0

The price of a European call option is given by

$$V(S, K, r, \sigma, T) = S\Phi(d_1) - Ke^{-rT}\Phi(d_2),$$

where Φ is standard normal CDF and d_1 and d_2 are defined as

$$d_1 = \frac{\log(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

and

$$d_2 = \frac{\log(S/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}}.$$

Digital Options

Digital or binary European option is a contingent claim of some underlying asset with

$$F(u) = \begin{cases} D, & u \geq K \\ 0, & u < K \end{cases}$$

for a call option and

$$F(u) = \begin{cases} 0, & u \geq K \\ D, & u < K \end{cases}$$

for a put option.

In the other words for a call option payoff is

$$F(S_T) = \begin{cases} 1, & S_T \geq K \\ 0, & S_T < K \end{cases}$$

and for a put option payoff is

$$F(u) = \begin{cases} 0, & S_T \geq K \\ 1, & S_T < K \end{cases}$$

A European digital call provides a payoff 1 unit (£,$) if the asset ends up above the strike price at the options maturity date, and 0 unit. otherwise.

$$
\begin{aligned}
V(S,T) &= De^{-RT}\mathrm{P}\left\{S_T \geq K\right\} = De^{-RT}\int_{Z_K}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-z^2/2}dz = \\
&= De^{-RT}\Phi(-Z_K),
\end{aligned}
$$

where

$$S_T = Se^{(R-\frac{\sigma^2}{2})T+\sigma W_T} \geq K$$

means

$$Se^{(R-\frac{\sigma^2}{2})T+\sigma\sqrt{T}Z} \geq K,$$

or

$$Z_K : Se^{(R-\frac{\sigma^2}{2})T+\sigma\sqrt{T}Z_K} = K,$$

$$Z_K = \frac{1}{\sigma\sqrt{T}}\log\frac{Ke^{-RT}}{S} + \frac{1}{2}\sigma\sqrt{T}.$$

Defining

$$d_2 = -Z_K = \frac{1}{\sigma\sqrt{T}}\log\frac{Se^{RT}}{K} - \frac{1}{2}\sigma\sqrt{T},$$

we conclude that for the European digital call

$$V(S,T) = C_{\text{binary}} = De^{-RT}\Phi(d_2),$$

and for the put

$$P_{\text{binary}}{=}De^{-RT}\Phi(-d_2).$$

also known as cash-or-nothing options: the pricing formula is

$$C_{\text{binary}}{=}De^{-RT}\Phi(d_2),\, d_2 = \frac{\log\frac{S}{K} + (R - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}},$$

$$P_{\text{binary}}{=}De^{-RT}\Phi(-d_2),\, d_2 = \frac{\log\frac{S}{K} + (R - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}.$$

$$C_{\text{binary}} + P_{\text{binary}} = De^{-RT}.$$

Example

Consider now a cash-or-nothing call and put options with the following parameter
valued:$S = \$100,\ K = \$105,\ T = 0.5$(years), $\sigma = 0.2,\ R = 0.1,\ D = \$100$ (the fixed cash amount that the option owner receives if the option finishes in-the-money).

For these values

$$d_2 = \frac{\log \frac{100}{105} + (0.1 - \frac{0.2^2}{2})0.5}{0.2\sqrt{0.5}} = -6.2156 \times 10^{-2} = -0.062156$$

$$\Phi(-0.062156) = 1 - \Phi(0.062156) = 1 - 0.5239 = 0.4761$$

Example contd...

$$C_{\text{binary}} = 100e^{-0.1\cdot 0.5}0.476\,1 = \$45.288$$

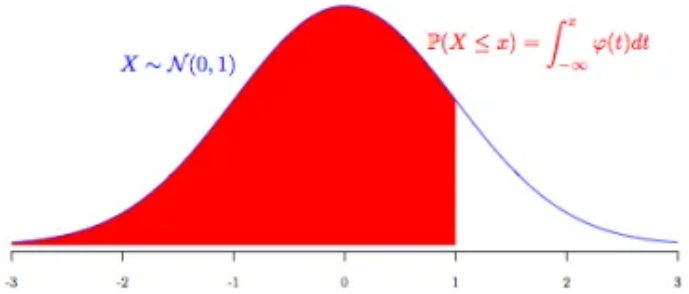
$$P_{\text{binary}} = 100e^{-0.1\cdot 0.5}0.5239 = \$49.835$$

$$C_{\text{binary}} + P_{\text{binary}} = De^{-RT}$$

$$45.288 + 49.835 = 95.123$$

$$100 \cdot e^{-0.1\cdot 0.5} = 95.123.$$

Normal CDF Table



	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Greeks

The option price sensitivities to different input parameters is called "Greeks".

1. The **Delta** of a derivative security Δ is defined as the rate of change of its price with respect to the price of the underlying asset. If the option price is C, then $\Delta = \frac{\partial C}{\partial S_0}$.
2. The second derivative of the call option with respect to the price of the stock is called the **Gamma** of the call option and is given by $\gamma = \frac{\partial^2 C}{\partial S_0^2}$.
3. The rate of change of the price of the call option with respect to time to maturity is called **Theta** and denoted by $\Theta = \frac{\partial C}{\partial T}$.
4. The **vega** of a derivative security denoted by $\wedge = \frac{\partial C}{\partial \sigma}$ is the rate of change of its price with respect to the volatility of the underlying asset.
5. The **rho** of a derivative security is the rate of change of its price with respect to the interest rate r and is denoted by $\rho = \frac{\partial C}{\partial r}$.

Let

$$W = (r - \sigma^2/2)t + \sigma\sqrt{t}Z.$$

Then W is normal random variable with mean $(r - \sigma^2/2)t$ and variance $t\sigma^2$. Let t represent the time to maturity. The Black-Scholes option pricing formula can be written as

$$C = \mathbb{E}[e^{-rt} I(se^W - K)],$$

where

$$I = \begin{cases} 1 & \text{if } se^W > K \\ 0 & \text{if } se^W \leq K. \end{cases}$$

Thus here

$$e^{-rt} I(se^W - K) = \begin{cases} e^{-rt}(se^W - K) & \text{if } se^W > K \\ 0 & \text{if } se^W \leq K. \end{cases}$$

We have

$$\frac{\partial}{\partial x} e^{-rt} I(se^W - K) = \begin{cases} \frac{\partial}{\partial x} e^{-rt}(se^W - K) & \text{if } se^W > K \\ 0 & \text{if } se^W \leq K. \end{cases}$$

Or, we can write $\frac{\partial}{\partial x} e^{-rt} I(se^W - K) = I \frac{\partial}{\partial x} e^{-rt}(se^W - K)$. Which further implies

$$\begin{aligned} \frac{\partial C}{\partial x} &= \frac{\partial}{\partial x} \mathbb{E}[e^{-rt} I(se^W - K)] \\ &= \mathbb{E}\left[\frac{\partial}{\partial x} e^{-rt} I(se^W - K)\right] \\ &= \mathbb{E}\left[I \frac{\partial}{\partial x} e^{-rt}(se^W - K)\right]. \end{aligned}$$

The partial derivative $\frac{\partial C}{\partial s}$ is called delta. Note that

$$\frac{\partial}{\partial s} e^{-rt}(S(t) - K) = e^{-rt}\frac{\partial S(t)}{\partial s} = \frac{S(t)}{s} e^{-rt}.$$

Thus,

$$\begin{aligned}
\frac{\partial C}{\partial s} &= \frac{e^{-rt}}{s}\mathbb{E}[IS(t)] = \frac{e^{-rt}}{s}\mathbb{P}(S(t) > K) \\
&= \frac{e^{-rt}}{s}\int_{z_1}^{\infty} s e^{(r-\frac{1}{2}\sigma^2)t+\sigma\sqrt{t}y}\frac{1}{\sqrt{2\pi}}e^{-y^2/2}dy \quad (\text{here } z_1 = \sigma\sqrt{t} - \omega) \\
&= \int_{z_1}^{\infty} e^{-\frac{1}{2}\sigma^2 t+\sigma\sqrt{t}y}\frac{1}{\sqrt{2\pi}}e^{-y^2/2}dy \\
&= \frac{1}{\sqrt{2\pi}}\int_{-\omega}^{\infty} e^{-u^2/2}du \quad (\text{by putting } u = y - \sigma\sqrt{t}) \\
&= \Phi(d1),
\end{aligned}$$

which is the change in the option price per unit change in the underlying price assuming all other parameters constant.

Example

Calculate the Delta for a 6-month call option on an asset that pays no dividends, assuming a volatility of 16%, an interest rate of 5% and a strike price is equal to 90% of spot.

Solution. $K = 0.9S$; $R = 0.06$; $\sigma = 0.16$; $T = 0.5$;
$d_1 = \frac{1}{\sigma\sqrt{T}} \log \frac{S}{K} e^{RT} + \frac{1}{2}\sigma\sqrt{T} = \frac{1}{0.16\sqrt{0.5}} \log \frac{1}{0.9} e^{0.06 \cdot 0.5} + \frac{1}{2} 0.16\sqrt{0.5} = 1.1348$,
$\Phi(d_1) = \Phi(1.1348) \simeq 0.8719$.This is for time $t = 0$.
For time $t \leq 0.5$

$$\Delta_{call} = \frac{\partial}{\partial S} C = \Phi\left(\frac{\log \frac{1}{0.9} + \left(0.05 + \frac{0.16^2}{2}\right)(0.5 - t)}{0.16\sqrt{0.5 - t}}\right) > 0.$$

Since $S(t)$ is independent of K, we have

$$\frac{\partial}{\partial K} e^{-rt}(S(t) - K) = -e^{-rt}.$$

Thus

$$\frac{\partial C}{\partial K} = \mathbb{E}[-Ie^{-rt}] = -e^{-rt}\mathbb{E}[I] = -e^{-rt}\Phi(d_2).$$

We have

$$\begin{aligned}\frac{\partial}{\partial r} e^{-rt}(S(t)-K) &= -te^{-rt}(S(t)-K) + e^{-rt}\frac{\partial S(t)}{\partial r} \\ &= -te^{-rt}(S(t)-K) + e^{-rt}tS(t) \\ &= Kte^{-rt}.\end{aligned}$$

Thus

$$\frac{\partial C}{\partial r} = Kte^{-rt}\mathbb{E}[I] = Kte^{-rt}\Phi(d_2).$$

Definition (Complete market)

A market is said complete if any contingent claim admits a replication strategy.

Theorem (First fundamental theorem of asset pricing)

A market model defined by $(\Omega, \mathcal{F}, \mathcal{F}_t, \mathbb{P})$ *and asset prices* $S_t,\ t \in [0, T]$ *is arbitrage free iff there exists a probability measure* $\mathbb{Q} \sim \mathbb{P}$ *such that the discounted asset prices* $(e^{-rt} S_t)$ *are martingales with respect to* $\mathbb{Q}$.

Theorem (Second fundamental theorem of asset pricing)

An arbitrage free market defined by the assets $(S_t^0, S_t^1, \cdots, S_t^d)$, *described as stochastic processes on* $(\Omega, \mathcal{F}, \mathcal{F}_t, \mathbb{P})$, *is complete iff there exists a unique martingale measure* $\mathbb{Q} \sim \mathbb{P}$.

Adding Jumps to GBM

Jumps in equity prices

- The GBM paths are continuous and hence, it can't model the jumps in the security's prices in either up or down direction.
- However such jumps do occur in practice.
- Next, we will consider such kind of model.

Modeling arrival of jumps

- We can model the number of jumps that occur by time t using a Poisson process $N(t)$ with intensity λ. The PMF of Poisson process is given by

$$\mathbb{P}(N(t) = n) = \frac{e^{-\lambda t}(\lambda t)^n}{n!}$$

- Let us suppose that, when the ith jump occurs the price of the security is multiplied by the amount J_i, where $J_1, J_2, \ldots$ are independent random variables having a common specified probability distribution.

- The GBM model with jumps will become

$$S(t) = S^*(t) \prod_{i=1}^{N(t)} J_i, \quad t \geq 0,$$

where $S^*(t)$ is GBM such that

$$S^*(t) = S^*(0) e^{(\mu - \sigma^2/2)t + \sigma B(t)}.$$

Further, the GBM is independent of J_i and the times at which the jump occurs.

Let

$$J(t) = \prod_{i=1}^{N(t)} J_i.$$

We have

$$\begin{aligned}
\mathbb{E}[J(t)] &= \mathbb{E}\left[\mathbb{E}\left[\prod_{i=1}^{N(t)} J_i | N(t)\right]\right] \\
&= \mathbb{E}[(\mathbb{E}J_i)^{N(t)}] = \sum_{n=0}^{\infty} (\mathbb{E}J_i)^n \frac{e^{-\lambda t}(\lambda t)^n}{n!} \\
&= e^{-\lambda t} e^{\lambda t \mathbb{E}J_i} = e^{-\lambda t(1-\mathbb{E}J_i)}.
\end{aligned}$$

- Let $X \sim N(\mu, \sigma^2)$, then
$$\mathbb{E}(e^X) = e^{\mu + \frac{\sigma^2}{2}}.$$

- Thus, we have
$$\begin{aligned}\mathbb{E}(S^*(t)) &= S^*(0)\mathbb{E}\left[S^*(0)e^{(\mu-\sigma^2/2)t+\sigma B(t)}\right] = S^*(0)e^{(\mu-\sigma^2/2)t+(\sigma^2/2)t} \\ &= S^*(0)e^{\mu t}.\end{aligned}$$

- Thus, security-buying bets will be fair bets i.e. $\mathbb{E}S^*(t) = S^*(0)e^{rt}$ if $\mu = r$.
- Hence, we take $\mu = r$ in the GBM model to make it a fair bet.

- We have

$$\mathbb{E}[S(t)] = \mathbb{E}[S^*(t)J(t)] = \mathbb{E}[S^*(t)]\mathbb{E}[J(t)] = S^*(0)e^{\mu t}e^{-\lambda t(1-\mathbb{E}J_i)}$$
$$= S^*(0)e^{(\mu-\lambda(1-\mathbb{E}J_i))t}$$

- Thus the security buying bets will be fair if

$$r = \mu - \lambda(1 - \mathbb{E}J_i)$$

or $\mu = r + \lambda - \lambda\mathbb{E}J_i$.

- Thus, the equity price process in risk neutral world will be

$$S(t) = S^*(0)e^{(r+\lambda-\lambda\mathbb{E}J_i-\sigma^2/2)t+\sigma B(t)}\prod_{i=1}^{N(t)} J_i.$$

No-arbitrage cost of a European call option

The no-arbitrage cost of a European call option having strike K and expiration time t is

$$c = \mathbb{E}\left[e^{-rt} \left(S^*(0) e^{(r+\lambda-\lambda\mathbb{E}J_i-\sigma^2/2)t+\sigma B(t)} \prod_{i=1}^{N(t)} J_i - K \right)^+ \right].$$

Merton Jump Diffusion Model

The equity price under Merton jump diffusion model is give by

$$S(t) = S(0)\exp\left((\mu - \lambda\kappa - \frac{1}{2}\sigma^2)t + \sigma B(t)\right)\prod_{i=1}^{N(t)} J_i,$$

where $N(t)$ is the Poisson process which counts the number of jumps till time t and J_i are iid log normal random variables such that $\log(J_i) \sim \text{Normal}(m, \delta^2)$ and $\kappa = \mathbb{E}(J_i - 1)$. Further, J_i, $N(t)$ and $B(t)$ are independent.

Volatility

Definition (Volatility)

Volatility is a statistical measure of the dispersion of returns for a given security. Volatility is generally measured as the standard deviation of returns from a financial security.

Facts about volatility

- Louis Bachelier used the term “coefficient of nervousness” or “of instability” of the price to represent volatility
- More volatility means generally more risk.
- Volatility estimation is very important for option pricing.

Definition (Historical Volatility)

Historical volatility of a security is the variance of the changes in the logarithm of the price of the underlying asset, obtained from past data.

Definition

Implied volatility of a security is the numerical value of the volatility parameter that makes the market price of an option equal to the value from the Black-Scholes formula.

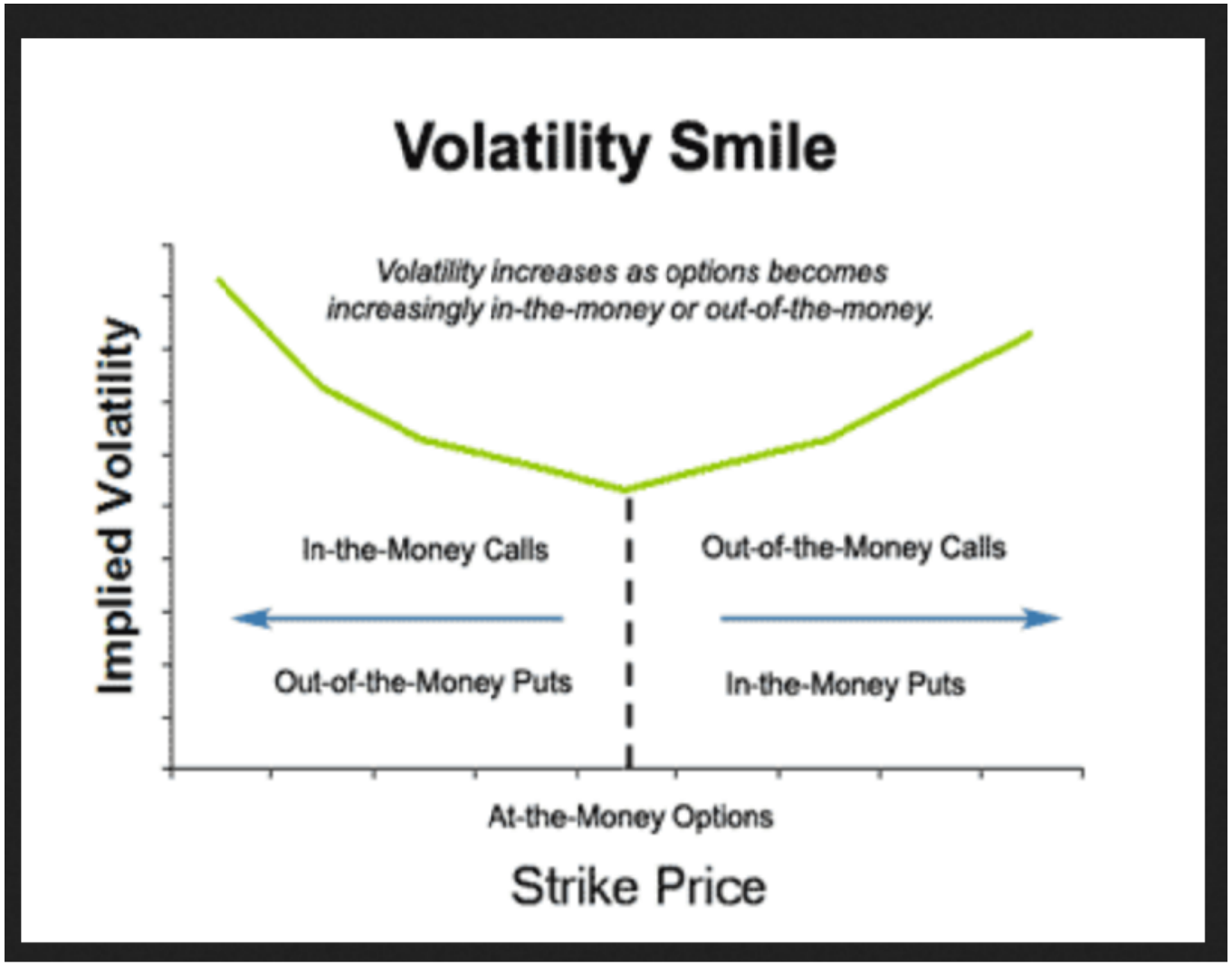


Figure: Volatility Smile

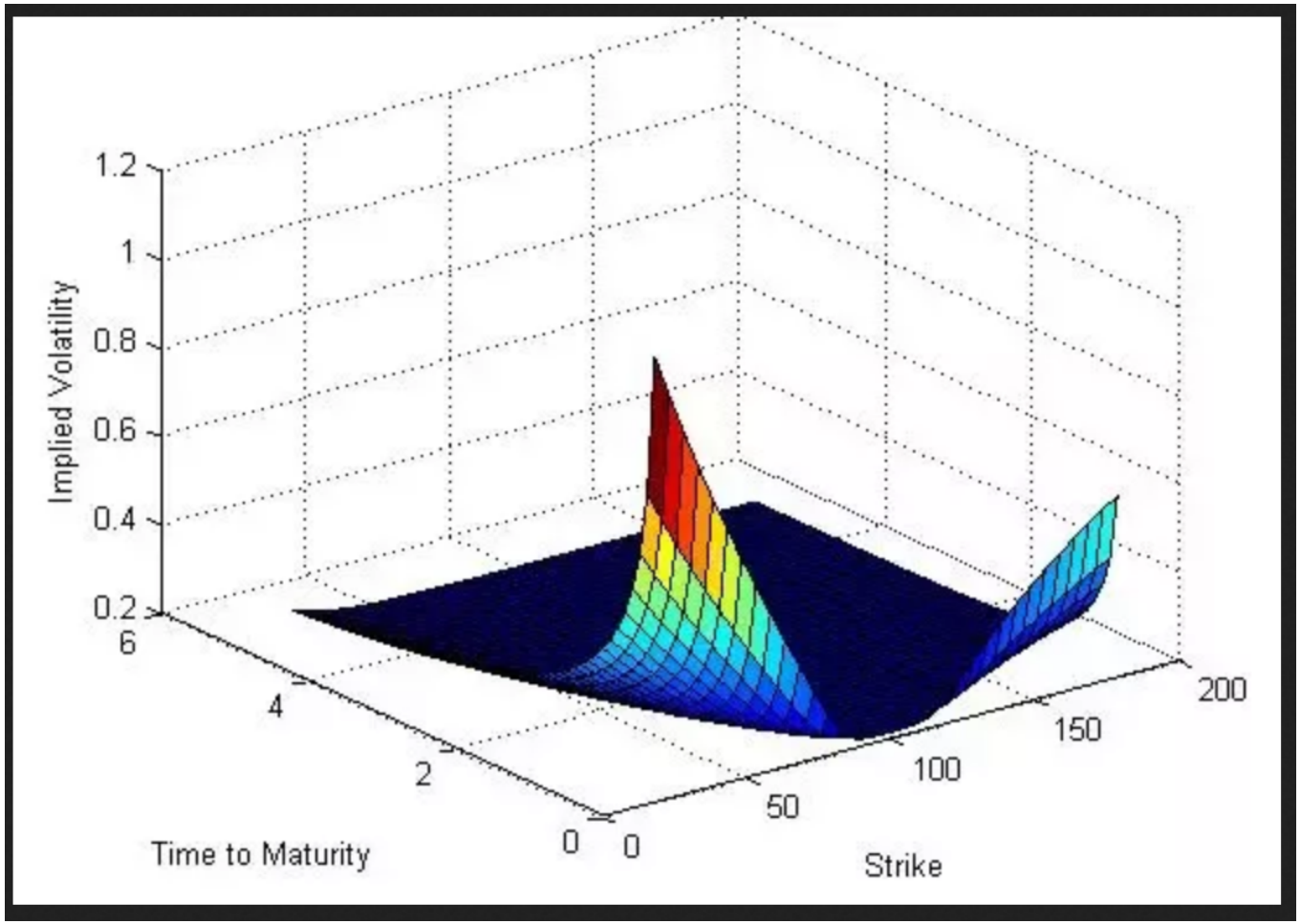


Figure: Volatility Surface

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THANK YOU